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## RESEARCH ARTICLE

# Sensory Substitution Device for Time Presentation: Spatial–Temporal Vibrotactile Encoding for Presenting Time on the Human Wrist

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**ABSTRACT** Presenting information privately such as alertness levels and time on the wrist via vibrotactile feedback proves invaluable for visually impaired individuals. Additionally, in situations where the visual channel is occupied, this serves as a discreet solution for sighted users, allowing them to stay informed during meetings or tasks without the need to overtly check their watches, thus minimizing potential distractions. However, it is a challenging task to present time accurately and efficiently to the users using vibrotactile modality due to the perceptual limits of human's haptic channel. Inspired by the metaphors of mechanical and digital watches that have been widely used in our daily lives, we proposed two novel spatial-temporal vibrotactile encoding strategies. By varying the location, number, and duration of the vibrotactile stimuli, these strategies are capable of presenting the exact information about the current time through a series of encoded tactile cues. A physical prototype was developed and fifteen participants were recruited to evaluate the two solutions. Two experiments were performed to evaluate the two encoding strategies. The results showed that the mechanical and digital encoding strategies achieved an average correct rate of  $90.55 \pm 5.2\%$  and  $95.22 \pm 4.1\%$  during the focused state, and  $95.28 \pm 3.3\%$  and  $97.78 \pm 3.8\%$  during the distracted state, respectively (*mean*  $\pm$  *SD*). Experimental results provide deep insights into utilizing the spatial-temporal patterns of vibrotactile stimuli for developing industrial-scale wearable haptic devices to present time and quantitative information efficiently and privately to the users.

**INDEX TERMS** Eye-free communication, haptic watch, spatial–temporal patterns, vibrotactile stimuli, wrist-worn tactile display, wearable computing, wearable haptic devices, information theory.

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## I. INTRODUCTION

Wearable haptic devices and wearable tactile displays are becoming popular in recent years [1], [2], [3], [4], [5],



visually impaired users however, it is difficult to use this watch for abled users who are not familiar with the standard Braille numbering system [19]. Ohtsuka et al. presented a vibration watch using a cell phone for visually impaired users. The only drawback is that it takes a long training time for participants to memorize the complex vibration patterns. Furthermore, as users normally do not hold the mobile phone all the time, therefore there is a huge chance of missing the vibration if the phone is being carried in pockets or bags [20]. Tam et al. developed a Wireless Wrist-worn Haptic Notification System (HaNS) to moderate time during oral presentations in conferences and seminars. Experimental results show the effectiveness of the system in reminding the speaker to manage time during oral presentations [21]. The meteor vibrating watch is a hand-held device with 3 buttons to represent hours, tens of minutes, and minutes respectively. The watch presents hours and minutes separately by using vibrotactile sensations in the form of short and long cues, with a long pulse for 5 units and 1 pulse for 1 unit. The watch can provide accurate time information, but users need to memorize the complex rules and also it is not a wearable device [22].

Besides presenting time information, haptic-based wrist-watches have also been developed to present other information [23]. Pasquero et al. implemented a haptic-enabled wristwatch with a single custom-made actuator to support eyes-free communication with a mobile device. Their focus was on enabling gesture-based active interactions in which the user gets information about the inbox of email by touching the watch. Their study indicated an overall accuracy of 73.6% for the identification of a delivered number of pulses. Moreover, participants responded positively to the informative tactile feedback [24]. Bosman et al. established a wearable wristband system comprising two vibrotactile devices. Each wrist was mounted with a single actuator. Their experimentation revealed that the system helped deliver directional information to the pedestrians. This system helped in reducing the number of errors to reach the destination [25]. Rekimoto et al. developed the Gesture Wrist, a gesture input device in the form of a wristband, a wrist-worn device that recognizes hand gestures and forearm movements [26]. Tsetserukou and Tachi developed BraTact, a tactile wristband comprising six symmetrically arranged vibration motors. Their objective was to convey the shape and stiffness of any colliding object teleoperation [27]. Weber et al. assessed VibroTac, a system similar to BraTact, for providing spatial guidance (translation and rotation of the hand) [28]. A wearable mood-reading device called the “Emo bracelet” was developed to communicate with the wearer using displayed light and vibrations [1]. Lee et al. explored a design space for the watch back tactile display using a  $3 \times 3$  tactators array and presented practical guidelines for the design of an efficient watch back tactile display [29].

Lee and Starner revealed that users could discriminate 24 tactile patterns with up to 99% accuracy after 40 minutes

of training. Among their adopted four parameters (intensity, starting point, temporal pattern, and direction) that vary in the 24 patterns, they found that intensity is the most difficult parameter to distinguish and temporal pattern is the easiest [30]. In all the above wearable haptic watches, the key technical challenge is the design of the vibrotactile encoding strategy. To present accurate and detailed information to the user, careful design of the key parameters need to be studied, including the number of vibrotactile cues, the layout and location of the cues on the wrist/arm, and the spatial-temporal pattern of the cues. For the time presentation requirement, the specific challenge is to provide accurate and efficient communication of current time to the user. Furthermore, the vibration pattern should provide the least cognitive workload for users to memorize diverse patterns, and require as short training time as possible. To the best of our knowledge, it is still an open problem of presenting accurate information of the current time (i.e. hour, minute, second) on the wrist using vibrotactile cues efficiently in real-world distracted environments.

## B. CONTRIBUTIONS OF THIS PAPER

Contributions of this paper can be summarized as:

- 1) We developed a novel filtering criteria that integrates biological constraints related to perceptual workload with three distinct types of cognitive demands: working memory, attention, and decoding workload. This comprehensive framework acts as a powerful tool for efficiently narrowing down large sets of potential solutions, allowing us to identify the optimal spatiotemporal patterns for conveying information via the vibrotactile modality.
- 2) We provided deep insights into various commercially available vibrators and their key characteristics, emphasizing the importance of selecting the appropriate actuator type for industrial-scale wrist-worn wearable devices. Additionally, we highlighted the hardware constraints, such as the coupling of frequency and amplitude in low-cost motors, which must be considered due to their significant impact on the accuracy and quality of vibrotactile cues.
- 3) We developed a physical prototype based on two encoding strategies, using metaphors of mechanical and digital watches. Evaluation experiments for presenting time in visually and auditorily distracted environments validated the performance of the selected solutions. An average correct rate of nearly 95% and an average reaction time of 10-12 seconds were achieved after a brief training period of 10 minutes.

## C. ORGANIZATION OF THIS PAPER

The remainder of this paper is organized as follows. In the remainder of Section II, we briefly summarize the design requirements and challenges for the haptic watch. In Section III, a filtering criteria set that consists of biological

**TABLE 2.** Summary of related work and comparison with the proposed study.

| Study                        | Focus/Contribution                                            | Limitations                                                                                                                                                                  |
|------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HapticaTM Braille Watch [19] | Time presentation using Braille system                        | Requires Braille literacy, limited to blind users, and not suitable for sighted users who may require discreet, non-visual feedback in busy or professional settings.        |
| Ohtsuka et al. [20]          | Vibration watch using cell phones                             | Long training time, non-wearable, and lacks portability, making it impractical for everyday use or continuous accessibility in dynamic environments.                         |
| Tam et al. [21]              | Haptic notification system for speakers                       | Limited to presentation reminders and lacks the versatility to convey diverse types of information such as time or contextual alerts.                                        |
| Meteor Vibrating Watch [22]  | Hand-held device with vibrotactile cues for time presentation | Users need to memorize complex rules for short/long pulses, and the device is not wearable, reducing practicality.                                                           |
| Pasquero et al. [24]         | Haptic-enabled smartwatch for emails                          | Focused on gesture-based interactions only, making it unsuitable for tasks requiring precise or continuous information delivery like time presentation.                      |
| Bosman et al. [25]           | Directional guidance using wristbands                         | Not suitable for time presentation, as it lacks the necessary encoding mechanisms to convey temporal information accurately and intuitively.                                 |
| Rekimoto et al. [26]         | Gesture wrist for motion detection                            | Gesture input focus, no time presentation, and does not provide tactile feedback specifically designed for conveying temporal or sequential data effectively.                |
| Tsetserukou and Tachi [27]   | BraTact tactile wristband for teleoperation                   | Focused on conveying shape and stiffness of objects, not optimized for presenting temporal or sequential data.                                                               |
| Weber et al. [28]            | VibroTac for spatial guidance                                 | Focused on hand translation/rotation, not designed for time presentation or minimizing cognitive workload.                                                                   |
| Lee et al. [29]              | Tactile display using 3x3 tactators array                     | Provided guidelines for tactile displays but not specifically designed for efficient time presentation.                                                                      |
| Lee and Starner [30]         | Discrimination of tactile patterns with high accuracy         | Requires 40 minutes of training for users to memorize patterns, with intensity being a challenging parameter to distinguish.                                                 |
| Proposed Study               | Spatial-temporal encoding for time                            | Efficient, private time presentation on the wrist, offering a discreet and intuitive solution suitable for both visually impaired and sighted users in various environments. |

constraints from four aspects of cognitive workload is proposed. In Section IV, we introduced a physical prototype based on two spatial-temporal encoding strategies based on mechanical and digital metaphors. Experimental results and discussions are introduced in Section V. Conclusions and future work are proposed in section 6.

## II. DESIGN REQUIREMENTS OF A HAPTIC WATCH

In this section, design requirements and challenges are summarized for the haptic watch. A systematic methodology is proposed to meet these requirements.

### A. REQUIREMENTS AND CHALLENGES

To present the current time through the vibrotactile cues on the wrist, there are four key challenges: accuracy of time information, the correct rate of user's response, reaction time to obtain the time information, and the mental workload for training.

First, the accuracy or the resolution of time presentation. The original requirement for presenting the time is to provide a fine second-level resolution (i.e. to tell the exact second of the current time) in a large period (24 hours). As the combinations of the current second are 86488 choices (i.e. 24 Hours x 60 Minutes x 60 Seconds), it seems difficult and impossible to communicate such huge information through a haptic perception channel. For some special application scenarios such as attending a meeting, the time resolution could be reduced to five minutes or one minute instead of one second,

and the covered period is 12 hours instead of 24 hours. The underlying assumption is that this watch will be mainly used in the daytime instead of nighttime, and the resolution of five minutes will provide sufficient information for people to have a rough idea about the current time during a meeting.

Second, the correct rate of the user's response to the presented cues. For a practical watch, the correct rate should be greater than 75%. The correct rate could be defined as the ratio between number of successful trials and the number of all trials:

$$C = \frac{\sum_{i=1}^N S_i}{N} \times 100 \quad (1)$$

where C stands for the correct rate,  $S_i$  denotes the number of successful trials and N denotes the total number of trials.

Third, the reaction time to obtain the time information is termed as the elapsed time including the presentation of sensory stimuli and the subsequent behavioral response according to the stimuli, or in other words we can term it as the processing speed of the sensory channel. Therefore, the reaction time includes time duration of the vibrotactile cues, and the time utilized by the participant to compute the time. However, it is very hard to present an accurate time including hours, minutes, and seconds within a very brief reaction time such as less than 1s. The reason is that this time perception process is a multi-processing task that involves vibrotactile perception, counting of cues, and mapping of the stimuli to the current time.

Last but not least, the practicality of the watch depends upon the mental workload. A haptic watch is a gadget that has to perform its function in highly demanding conditions when the user is attending to or focusing on another task. The patterns of vibrotactile cues should be easily rememberable and perceptible by the user. The complexity of the temporal patterns induces a high mental workload. Theoretically speaking, complex vibrotactile patterns might encode detailed time information, however, more complex vibrotactile patterns might introduce a larger mental workload and require longer training time for users to get familiar with the vibrotactile patterns.

## B. METHODOLOGY

To meet the above challenges, we proposed a methodology (Fig. 1). Based on the identified requirements of the haptic watch, we proposed filtering criteria that consisted of perception workload and three aspects of cognitive workload: working memory, attention, and encoding workload. The hardware constraints refer to the capability of the selected low-cost ERM motors.

Two encoding strategies were proposed and implemented with inspiration from the metaphors of mechanical and digital watches. A physical prototype was developed and fifteen participants were recruited for the evaluation of the two strategies. Two experiments were performed in either a focused state or a distracted environment.

## III. FILTERING CRITERIA

In this section, a set of filtering criteria based on perceptual constraints and hardware constraints is proposed for selecting the best possible encoding strategy to accurately and efficiently present current time to the human wrist.

The cognitive process of estimating current time using vibrotactile stimuli includes four cognitive blocks: perception, working memory, computation, and sustained attention (Fig. 1). In case of external disturbance, selective attention is also needed to guide the attentional focus to the vibrotactile cues at the needed time. Based on the above cognitive process, the filtering criteria could be constructed based on the perceptual, cognitive, and hardware constraints.

### A. PERCEPTUAL CONSTRAINTS

Perception of the vibrotactile cues relied on the type and the parameters of the cue. These include frequency, intensity, duration, contact area, and inter-stimulus interval. Based on the selection of these parameters of the vibrotactile stimuli the perception is determined. Understanding of human detection threshold and discrimination threshold plays an important role in ensuring reliable perception. Several studies have already been conducted on analyzing and selecting these parameters for the vibrotactile stimuli [31], [32], [33], [34].

#### 1) FREQUENCY

Numerous studies have been conducted to study the discrimination and the detection thresholds of frequency. The

detection threshold for the frequency of vibrotactile stimuli ranges from 0.4-1000Hz [35]. The human skin is most sensitive to frequencies ranging from 150-300Hz. However, these value varies for the site of the stimuli. For the wrist, the maximum sensitivity occurs at 220Hz [36]. Therefore, it is suggested to choose this range of frequencies while designing wrist-worn wearable devices.

In addition to the vibrotactile signal with a constant frequency, frequency-modulated signals are also widely used in haptic devices. Phong et al. suggested in their study that frequency modulation-based vibrotactile feedback is more useful than the signal with a constant frequency [37].

#### 2) AMPLITUDE

Craig suggests that to discriminate the two sinusoidal stimuli the intensity difference between the two stimuli should be more than 3dB [38]. Choi et al. suggested that at least 20%-30% of a difference in amplitude or frequency is necessary for the robust discrimination between vibrotactile stimuli in practical applications [34]. Verillo suggested that the sensation of the vibrotactile stimuli decreases with the increase in the amount of energy applied to the skin [39]. The minimum threshold (smallest detectable displacement) is usually observed between 150 and 300Hz, and it can be smaller than  $0.1\mu\text{m}$ . This absolute threshold is affected by several factors, such as body site, contact area, stimulus duration, stimulus waveform, contact force, skin temperature, the presence of other masking stimuli, and age [40]. The standard amplitudes suggested by most of the literature are 200, 300, and  $400\mu\text{m}$ . However, the minimum JND is  $0.1\mu\text{m}$  [34].

Amplitude-modulated vibrotactile stimuli are widely used in the development of tactile displays and the tactile rendering domain. Park et al. suggested that the pulse-like low-frequency sensation of an amplitude-modulated vibration increased for very low modulation frequencies (1-10Hz), then decreased for higher modulation frequencies (10-80Hz), and eventually converged to the smooth vibrational sensation of the 150Hz carrier signal. They also suggested that the envelope waveform is primary information for the discrimination of amplitude-modulated signals, instead of their spectral energy distributions [41]. Amplitude-modulated vibrotactile signals are also influenced by the adaptation effect. Francisco et al. evidenced that faster modulation rates would lead to the stimulus being less adapted [42]. Stepp and Matsouka evidenced that amplitude modulation vibrotactile stimuli provide superior feedback for the object manipulation task than the pulse train frequency modulation stimulation. Amplitude-modulated stimulation provides a graded stimulus to replace the expected graded force feedback. Conversely, pulse train frequency modulation provides a discrete stimulation pattern that is quite different temporally from the expected force feedback. During amplitude modulation, the graded amplitude of the vibrotactile information is more like

the original signal, so may be integrated more effectively for the object manipulation task [43].

### 3) WAVEFORM

Hsu et al. suggested that comparatively skin is not as sensitive to the waveforms as the ear which is exquisitely sensitive to timbre [44]. Generally, it is not significant to design patterns by using more than one waveform while designing wearable tactile displays because it is difficult to discriminate the different waveforms.

However, some researchers still explored this domain. Bensmia et al. evidenced while studying complex vibrotactile waveforms consisting of two superimposed sinusoids at varying phases that it is easier to discriminate low-frequency (10Hz+30Hz) waveforms than the high-frequency (100Hz+300Hz) waveforms [45]. Chancey et al. suggested that sinusoidal and square waves tended to prompt quicker response time (RT) than the noise wave. A potential reason for this is likely because this waveform was less uniform in frequency [46]. According to most of the literature available, mostly the sinusoidal wave is utilized for developing the vibrotactile patterns.

### 4) DURATION AND ISI

The ISI should be selected at least greater than 20-30ms. Previous studies show that the ISI is a very vital parameter that plays a significant role in the perception of vibrotactile stimuli. The minimum ISI that a human can perceive between the two vibrotactile cues is 5ms [35].

The human tactile channel possesses high temporal acuity, e.g. A human can distinguish successive pulses with a time gap of 5ms [35]. This tactile sensitivity is better than for vision (25ms) but worse than auditory acuity (0.01ms) [47]. Kaaresoja et al. suggested that the preferred duration for the vibrotactile stimuli is between 50 to 200ms [48]. However, Hirsh and Sherrick suggested that the temporal order judgments cannot be made until the ISI is 20-30ms [49].

### 5) LAYOUT AND LOCATION

Studies reveal that spatial and temporal patterns are easier to discriminate than frequency and intensity [50], [51]. If the two tactators are placed very close to each other the participant will miss the stimuli and will perceive them as one location. This two-point discrimination threshold varies with the location of the stimuli on the body. Cholewiak et al. revealed that the two-point discrimination threshold is 35mm for the placement of tactators on the wrist [52]. The two-point discrimination experiment by Weinstein suggests that there should be a minimum distance of 38mm between the two stimuli to be properly differentiated on the left forearm [18]. As the width of a typical watch is about 34mm, which is less than the two-point discrimination threshold, therefore, the placement of multiple tactators on the same side of the wrist is not feasible therefore, we suggested a circumferential layout.

Generally, the dorsal side is utilized in the case of tactator placement in wristwatches. Very rarely volar and other sides are utilized. Studies revealed that the dorsal and the volar sides are comparatively more sensitive in vibrotactile perception as compared to the other two sites [53].

### 6) COORDINATION PATTERN

Zhang et al. also suggested that better performance can be achieved if the stimuli are presented sequentially [42]. The sequential temporal pattern produces significant results as compared to the simultaneous presentation [54]. Jones et al. evidenced that it is easier to identify the tactile patterns presented in a 3-by-3 array of motors mounted on the forearm if the motors were activated sequentially in a transverse (mediolateral) direction than in the longitudinal (distal-proximal) direction, even though the distance between the motors was the same in both directions [55]. Oakley et al. reported that when the vibrotactile stimuli are presented simultaneously at close distances, the perception is a single stimulus with a greater amplitude. The vibrotactile patterns that involve both directions of motor activation were even harder to identify [56]. Wang et al. explored simultaneous activation of the vibrotactile cues on the forearm and evidenced that the rate of correct localization of vibrotactile cues decreases linearly with the increase in the number of activated cues [57].

## B. COGNITIVE CONSTRAINTS

### 1) ATTENTION

Attention involves all the steps in the cognitive process of time presentation (Fig. 1). Both selective attention and sustained attention are required to support perception, working memory, and the encoding process. Multiple task performance simultaneously and successively in humans is based on their approach to managing attentional resources (selective, divided, or focused attention). When an individual is presented by multiple stimuli, the verdict to select, rank, or overlook the stimuli is mostly affected by the consistency of the information or the user's ability to process attentional phenomena (automatic or control processing). In general, the limited capacity of human attention persuades the selection of one task at the cost of other stimuli or time-sharing. When designing multitasking-friendly mobile UIs, exploring the ability to manage attentional resources is essential to ensure efficient interaction [58]. Rubinstein's [59] and Schumacher's [60] studies revealed that people's ability to perform dual tasks depends on resource management for stimulus-response (S-R) channels. The result of Schumacher's study indicates that humans can perform dual tasks to some extent when the S-R channel for one task does not overlap with the S-R channel for the other task. Thus, touch may provide an appropriate alternative modality to present information in mobile UIs when users have to reserve their visual attention for other tasks.

## 2) WORKING MEMORY

Working memory (WM) is the ability to maintain and manipulate information in the short term and the context of concurrent processing or distraction [61]. Working memory is a key cognitive process that influences the performance of wearable tactile displays. To achieve better performance, it is needed to design encoding strategies that induce a low working memory load that can be handled within human working memory capacity. While there are many works on verbal and visual-spatial working memory, there are few works on vibrotactile working memory capacity. As suggested by Bancroft et al. vibrotactile working memory shares common qualities with the other working memory systems [62]. Bancroft and Gamble also suggested that stored items of vibrotactile working memory may suffer from high levels of interference in the form of overwriting or representation-based interference. Furthermore, used a two-item delayed match-to-sample paradigm to demonstrate that subjects can store multiple items in vibrotactile working memory [63].

## 3) DECODING

After perceiving the vibrotactile cues, the user needs to translate them into the value of hour and minute based on a pre-defined encoding strategy. We defined this process as the decoding or information mapping. Different encoding strategies may lead to varied decoding workloads. In previous studies, typical encoding strategies for mapping stimulus to information include direct mapping and abstract encoding [64]. The latter involves sophisticated coupling between perceived stimulation and meaningful information whereas the former refers to an intuitive understanding of the stimuli through the combination of various parameters. One of the most well-known abstract encodings is Braille [65]. When developing solutions for wearable tactile displays to present information to the vibrotactile modality, a low decoding workload should be considered to enhance performance.

## C. HARDWARE CONSTRAINTS

When designing wrist-worn wearable devices, one important parameter to be considered is the type of the vibrator. All the quantitative parameters of the vibrotactile cues are dependent upon the accuracy and quality of the vibrators. Below we discuss some commercially available actuators that can be considered. Solenoids are widely used in tactile displays but the precise start and stop time is difficult to achieve with the solenoids. Engineering Acoustics, Inc., (EAI, Casselberry, FL) created the C2 and C3 actuators for the direct application of vibrations on the skin. Tactile Labs, Inc. (Montreal, C, Canada) designed the Haptuator for embedding into handheld tools and devices. Its housing possesses a diameter of 14 mm and a length of 29mm. This actuator can produce various waveforms and patterns by modulating key parameters such as frequency, amplitude, etc. However, it is not sufficiently compact to be integrated into a

**TABLE 3. Filtering criteria for designing encoding strategies using vibrotactile stimuli.**

| Aspect                 | Parameter               | Suitable Range                                                                    |
|------------------------|-------------------------|-----------------------------------------------------------------------------------|
| Perception constraints | No. of cues             | 2 simultaneous active cues<br>Sequential is better than simultaneous presentation |
|                        | Location                | Dorsal side is most sensitive<br>Minimum distance: 35mm                           |
|                        | Frequency               | 150-300Hz                                                                         |
|                        | Amplitude               | Envelope waveform                                                                 |
|                        | Waveform                | Sinusoidal wave                                                                   |
|                        | Duration                | 50 to 200msec                                                                     |
| Cognitive constraints  | ISI                     | > 20-30msec                                                                       |
|                        | Working Memory workload | Grouping information by chunks                                                    |
|                        | Attention workload      | Small sustained attention workload<br>Selective attention                         |
|                        | Decoding workload       | Mapping rule should be simple                                                     |
| Hardware constraints   | Pancake motor           | Coupling between frequency and amplitude                                          |

wristwatch [66]. Precision Microdrives (London, U.K.) offers LRAs (Linear Resonance Actuators) with 10mm diameter and 200Hz frequency for the vibrotactile displays. Lastly, ERMs (Eccentric rotating mass) have been implemented in a wide variety of vibrotactile displays because of their small size and robustness. ERMs are further classified into two types: shafted motors or cylindrical vibrating motors, and shaftless vibrating motors enclosed in a capsule often termed pancake or coin vibration motors. Precision Microdrives, Kotl, and other manufacturers offer a wide inventory of these ERMs [67], [68]. For further classification of actuators can be found in [34]. For some low-cost motors such as coin motors, the frequency, and amplitude are coupled and cannot be independently modulated. This hardware constraint needs to be taken into account when designing vibrotactile encoding solutions.

## D. PREFERRED SOLUTIONS

Based on the above knowledge of perceptual, cognitive, and hardware constraints, we summarized several typical filtering criteria that could help designers to determine the suitable range of key parameters of vibrotactile cues (Table 1). These criteria should be followed when designing feasible encoding strategies for time presentation on a wrist.

Based on the previous psychophysics studies in haptics perception and cognition, following important lessons could be summarized in addition to the above filtering criteria:

- 1) The number of cues is a reliable indicator, which can lead to robust discrimination performance. In contrast, human's discrimination capability on the following four parameters, including frequency, amplitude, waveform, and ISI is not as good as the discrimination capability on the number of cues. Therefore, the number of cues is a preferred variable for designing encoding strategies.
- 2) Based on human constraints of perceiving multiple cues, simultaneous vibration of multiple cues should be used in a cautious way as it might lead to a higher perceptual and cognitive workload than the sequential presentation of the stimuli.
- 3) Reducing the number of motors might increase the efficiency and the perceptual workload of the task. While designing a spatial-temporal strategy, it is preferred to utilize a minimum number of tactators and more information should be encoded in the temporal patterns.

- 4) When designing the temporal patterns it is better to reduce the number of vibrotactile cues to reduce the decoding or computation workload. This may enhance the performance and also reduce the reaction time.
- 5) It is better to reduce the working memory workload to avoid data loss. One typical solution is to divide the large information needed to be memorized into several small chunks and thus reduce the working memory workload.

#### IV. TWO ENCODING STRATEGIES OF HAPTIC WATCH

Based on the filtering criteria, we proposed two encoding strategies to develop a solution for an eye-free communication watch that can provide a relatively high time resolution in a limited time span. The underlying inspirations are from the two metaphors of the wrist watch i.e. mechanical and digital wrist watch.

##### A. MECHANICAL ENCODING STRATEGY

The first encoding strategy is designed based on the metaphor of typical mechanical watches. Supposing the cross-section of the wrist as a virtual dial plate of a watch, the vibrotactile cues aim to provide the location of the hour hand and the minute hand of the watch on the virtual plate (Fig. 2). In this strategy, we utilized low-cost coin vibration motors for presenting tactile stimuli. Therefore, the amplitude and the frequency can not be controlled individually. The frequency selected was 220Hz which is suggested to be the best frequency range for the wrist [36]. We utilized two vibration durations of 1 second and 500 msec for presenting hours and minutes respectively. The durations were intentionally set large for the effective perception of the stimuli much larger than the JND.

Twelve-hour format of time and 5 minutes resolution could be presented by this strategy. 4 vibration motors are mounted in a circumferential layout around the wrist. Each motor is assigned to represent one-quarter of the virtual dial plate. As shown in Table 2, in this mechanical encoding strategy, human discrimination of three parameters (location, duration, and number of cues) was combined to produce reliable discrimination between the 12 values for the current hour, or the 12 values for the current segment of minute. For each motor, the information of hours is presented by long-duration vibrations while minutes are presented by small-duration vibrations. For the adopted low-cost coin vibration motors, the long duration leads to a strong sensation, while the short duration leads to a weak sensation.

For representing twelve hours, the role of these quarters are defined as follows: motor-1 displays the range from the 1st to 3rd hour, motor-2 displays the range from the 4th to 6th hour, motor-3 displays the range from the 7th to 9th hour, and motor-4 displays the range from 10th to 12th hour. Similarly, for presenting minutes, we divided into four quarters and each motor is assigned to one quarter. Motor 1 displays the range from the 1st to 15th minutes, Motor 2 displays the range from the 16th to 30th minutes, Motor 3 displays the range from 31st to 45th minutes, and Motor 4 displays the range from 46th

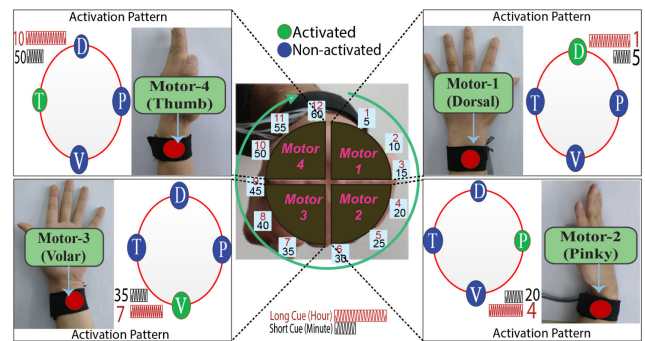


FIGURE 2. Motors location and the mechanical encoding strategy for the time presentation (Mechanical metaphor).

TABLE 4. Vibrotactile encoding strategy for the time (HR = Hour, MIN = Minutes, M = Motor).

| Motor No. | Long duration vibration for presenting Hours |          |            | Short duration vibration for presenting Minutes |           |            |
|-----------|----------------------------------------------|----------|------------|-------------------------------------------------|-----------|------------|
|           | One cue                                      | Two cues | Three cues | One cue                                         | Two cues  | Three cues |
| M-1       | 1st HR                                       | 2nd HR   | 3rd HR     | 0-5 MIN                                         | 5-10 MIN  | 10-15 MIN  |
| M-2       | 4th HR                                       | 5th HR   | 6th HR     | 15-20 MIN                                       | 20-25 MIN | 25-30 MIN  |
| M-3       | 7th HR                                       | 8th HR   | 9th HR     | 30-35 MIN                                       | 35-40 MIN | 40-45 MIN  |
| M-4       | 10th HR                                      | 11th HR  | 12th HR    | 45-50 MIN                                       | 50-55 MIN | 55-60 MIN  |

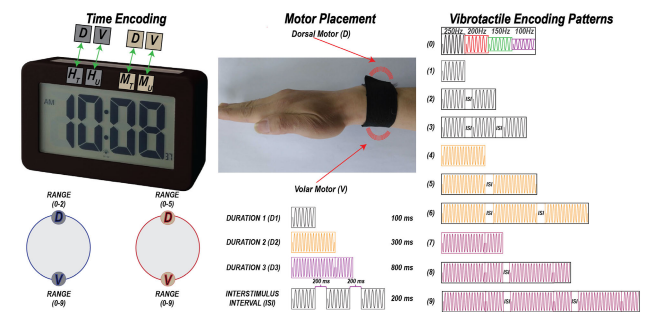


FIGURE 3. Digital encoding strategy for the time presentation (Digital metaphor).

to 60th minutes. We first presented this idea in ICMA as a conference paper [69]. In this paper, we further extended and improved the resolution of the time for the haptic watch and also presented a more rigorous comparison between the two strategies and psychophysical metrics as well.

The complete sequential vibration pattern of the motors is shown in Table 2. The temporal pattern selected to present the time is the sequential presentation of the stimuli, i.e., the hours and the minutes are presented sequentially one after the other. This pattern consists of vibration segments for the current hour followed by distinctive vibration segments for the current minute span (in five-minute resolution). The adopted combinations of the segments were easy to memorize. The goal of this design is to create uniqueness in each segment and to avoid possible confusion during the perception of time cues.

##### B. DIGITAL ENCODING STRATEGY

The second proposed strategy, the digital encoding strategy, is designed based on the metaphor of typical digital watches. The four digits of the current time are encoded using the

vibrotactile stimuli. Twenty-four-hour format of time and 1 minute resolution could be presented by this strategy.

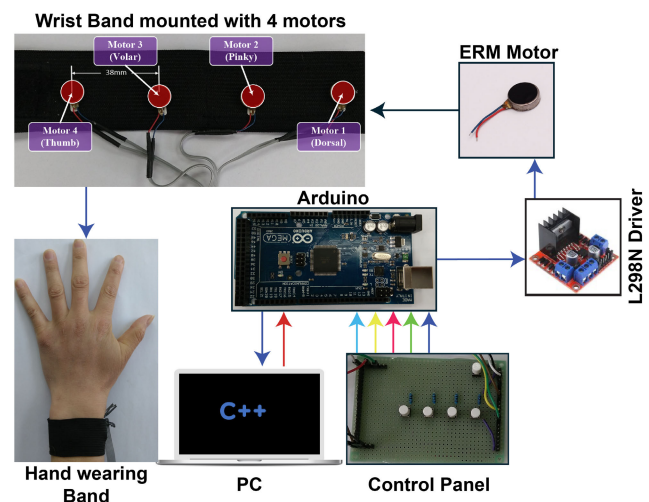
To reduce the working memory workload, sequential presentation of the four digits was adopted, and one break was introduced between the estimation of the hour and the minute. Benefitted by this design, the four digits were divided into two chunks estimating only two digits in each chunk. Two motors are utilized to present the two digits in each chunk. In the first chunk, the user perceived vibrotactile cues from M-1 and M-3 to estimate the first and second digits for the current hour respectively. After he/she was confident about the estimation results of the current hour, he/she pressed a button to trigger the vibration of M-1 and M-3 to perceive the cues for the two digits of the current minute.

Each digit has a different range of values (Fig. 3). For the first digit of the hour, the range is from 0 to 2, while the range is from 0 to 9 for the second digit of the hour. Similarly, the range is from 0 to 5 for the first digit of the minute, while it is from 0 to 9 for the second digit of the minute. Based on the filtering criteria, human discrimination of two parameters (duration and number of cues) was adopted to produce nine reliable temporal patterns for digits from 1 to 9. Specifically, the digit 0 was represented by a decayed vibration signal with four varied frequencies.

To implement this strategy, a pilot study with three participants was conducted to determine the values of two parameters in the encoding strategy: duration and the inter-stimulus interval between the two successive vibrotactile cues. For the vibration durations, three identifiable levels were selected as 100ms, 300ms, and 800ms respectively. The criteria for selecting these parameters were based on the performance of the participants. The ISI selected for all the digits was 200ms.

### C. PHYSICAL PROTOTYPE OF THE HAPTIC WATCH

The prototype of the watch comprises a Velcro-attached flexible wristband with vibration motors placed circumferentially to provide spatially distributed haptic feedback that serves as the foundation for its hardware components. Embedded within the wristband are four vibration motors, positioned 38mm apart, and four coin motors (Precision microdrives 310-103) with a diameter of 10mm. The placement of the vibration motors ensures optimal tactile perception on the wrist. The control electronics of the watch involves an Arduino ATmega 2560 microcontroller, which is responsible for managing the activation of the vibration motors. Pulse Width Modulation (PWM) is used to fine-tune vibration intensity and frequency. To facilitate the control of the motors, the system integrates two L298N H-Bridge drivers that regulate the amplitude and frequency of the coin vibration motors. For user interaction, the haptic watch features three push buttons. These buttons allow users to initiate time presentation and adjust settings. Button-1 (located on the extreme right) serves to start the timer and present hour



**FIGURE 4.** Illustration of hardware used for the prototype of the haptic watch and the control sequence and control panel.

stimuli, while Button-2 (in the middle) is designated for minute stimuli presentation. Button-3 (at the back) is used to stop the timer and record the user's reaction time. The software component of the haptic watch is developed in Microsoft Visual C++, which generates synthetic target times at random intervals. All data processed during the interactions, including button presses and reaction times, is recorded in a CSV format for analysis. This comprehensive integration of hardware and software components ensures an effective and intuitive user experience for the haptic watch (Fig. 4).

## V. EXPERIMENTS

### A. PARTICIPANTS

15 individuals (2 females, ages 22-30) participated in the experiment. The participants have an average wrist circumference of 15.43cm. All the participants were the graduate students of Beihang University. All the participants gave written consent to participate in the experiment. All were tested with their left arm wrist.

### B. EXPERIMENTAL DESIGN

Two experiments were conducted to evaluate the two strategies respectively. Each experiment is further divided into three blocks: a training block, a focused state testing block, and a distracted state testing block. To account for the counterbalance effect, the participants were randomly divided into two groups with equal numbers. The first group performed the mechanical encoding strategy first, and after three days performed the digital encoding strategy. The second group performed the two experiments in the reverse order.

The participants were asked to wear the wristband and rest their left arm on the table. The response of the current hour, minute, and recording buttons were provided by the participant's right hand (Fig. 4).

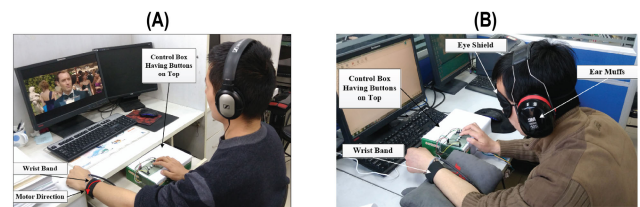
## 1) EXPERIMENT 1

The training block took about 10 minutes and was further divided into two phases, i.e. the basic training and advanced training phases. During the basic training phase, each motor on the wristband was sequentially actuated to help participants become familiar with their spatial locations. Vibrations with long or short durations were presented to enable the participant to differentiate the two durations. This phase lasted 5 minutes. It was followed by the advanced training phase comprised of 25 trials. The purpose of the advanced training phase was to enable the participant get familiarized with the encoding pattern of the mechanical watch metaphor. In each trial, the participant was presented with a random target time, and he/she needed to estimate the hour and minute based on the vibrotactile cues. After each response, the actual answer of the current time was provided to the participant for immediate feedback. The advanced training phase lasted for about 5 minutes. After completing the training block, participants were asked to take a rest of 5 minutes to avoid any numbness and adaptation. The formal experiment started from the first testing block. During the whole focused state testing block, the eyes of the participants were covered by the eye shield, and the ears were covered by ear muffs in order to avoid audio-visual distraction (Fig. 5 (A)). The participants were instructed to remain focused during the whole experiment. Thirty trials were presented during the focused state testing block.

The trial starts when the participant presses the push button 1 to start the trial and timer to record the reaction time. The cues for presenting hours were triggered when button 1 was pressed. After perceiving the vibrotactile stimuli, the participant had to respond to his/her estimation verbally to the experimenter. Immediately after reporting the hour information, the participant pressed button 2 to trigger the cues for presenting the minutes. After the participant finishes perceiving the minutes he/she had to immediately push the record button to record the total reaction time. After perceiving the time the participant had to respond verbally to the experimenter as quickly as possible, and the experimenter recorded the answer. After providing the answer to the experimenter, the participant had to immediately push the record button to record the reaction time. No feedback was provided to the participants regarding the correct rate and the response time during the experiment.

The total time of the focused state testing block was about 10 minutes. After the block, the participant was required to take a rest of 5 minutes to avoid fatigue and adaptation to the vibrotactile stimuli.

The distracted state testing block was comprised of 30 trials. In this block, an audio-visual distraction is introduced by playing a Hollywood comedy movie “SPY” on the monitor in front of the participants. The procedure for conducting the experiment and the instructions to the participants were the same as in the focused state testing block, except that the participant had to look at the screen



**FIGURE 5. Illustration of the experimental setup in the focused state testing block, Illustration of the experimental setup in the distracted state testing block.**

and watch the movie during the whole block (Fig. 5(B)). The total time for this block was 10 minutes.

At the end of each testing block, every participant was asked to fill out a questionnaire to express their feelings and opinions regarding the experiments.

## 2) EXPERIMENT 2

Experiment 2 is also divided into three blocks i.e. a training block, a focused state testing block, and a distracted state testing block. The training block took about 10 minutes and was further divided into two phases, i.e. the basic training and advanced training phases. In the basic training phase, the vibration pattern of each of the ten digits was provided on the wrist and the encoding pattern for each digit was explained to the participants. This phase lasted 5 minutes. It was followed by the advanced training phase comprised of 25 trials. In each trial, the participant was presented with a random target time, and he/she needed to estimate the hour and minute based on the vibrotactile cues. The correct answer to the trial was provided immediately after the user's report. The majority of the participants responded that after 25 trials training, they were able to identify the presented time easily. In this encoding strategy, the most important task was to exactly identify the temporal pattern and map the pattern to the current time. After the training session participants were instructed to take a rest of 5 minutes to avoid fatigue.

The focused state testing block starts after the training block. In this block, a total of 30 trials were presented to the participants. The trial starts when the participant presses the push button 1 to start the trial and timer to record the reaction time. The cues for presenting the two digits of the current hour were triggered when button 1 was pressed. After perceiving the vibrotactile stimuli, the participant had to respond to his/her estimation verbally to the experimenter. Immediately after reporting the hour information, the participant pressed button 2 to trigger the cues for presenting the two digits of the current minutes. After the participant finishes perceiving the minutes he/she had to immediately push the record button to record the total reaction time. During the experiment, participants were instructed to remain focused and pay attention to the stimuli. To avoid distractions from the environment, the eyes of the participants were shielded by the eye shield, and the ears were covered by the ear muffs to prevent audio-visual distraction (Fig. 5(B)).

**TABLE 5. Results of the participants during the experiment 1 (C.R. = Correct Rate, R.T. = Mean of reaction time S.D. = standard deviation of reaction time).**

| S.No.   | Focused |          |      | Distracted |          |      |
|---------|---------|----------|------|------------|----------|------|
|         | C.R.    | R.T. (s) | S.D. | C.R.       | R.T. (s) | S.D. |
| 1       | 100.00% | 6.47     | 1.35 | 96.66%     | 10.72    | 2.05 |
| 2       | 96.66%  | 8.48     | 1.16 | 96.66%     | 8.83     | 1.61 |
| 3       | 96.66%  | 10.12    | 1.41 | 93.33%     | 9.71     | 1.93 |
| 4       | 96.00%  | 8.27     | 1.22 | 93.33%     | 10.65    | 3.81 |
| 5       | 96.66%  | 8.55     | 1.27 | 90.00%     | 9.29     | 2.15 |
| 6       | 96.66%  | 10.51    | 2.34 | 93.33%     | 10.53    | 2.01 |
| 7       | 96.66%  | 13.98    | 8.89 | 86.66%     | 16.8     | 5.48 |
| 8       | 96.66%  | 10.47    | 2.82 | 86.66%     | 10.51    | 2.30 |
| 9       | 83.33%  | 10.84    | 2.41 | 80.00%     | 12.01    | 4.82 |
| 10      | 93.33%  | 8.48     | 1.53 | 86.66%     | 9.70     | 2.04 |
| 11      | 96.66%  | 8.61     | 1.83 | 96.66%     | 10.18    | 2.62 |
| 12      | 93.33%  | 10.47    | 3.39 | 86.66%     | 14.19    | 2.61 |
| Average | 95.22%  | 9.60     | 2.47 | 90.55%     | 11.09    | 2.79 |

The distracted state testing block starts after 5 minutes of rest. In this block, again the same Hollywood movie “SPY” was played on the monitor. This block also comprises 30 trials. The procedure for the rest of the experiment is the same as that for the focused state testing block. The total time for this block is about 10 minutes.

### C. RESULTS

The prototype of the watch was evaluated based on three fundamental parameters, i.e. correct rate, reaction time, and information transfer rate, which aim to quantify the accuracy and efficiency of the proposed prototype of the watch.

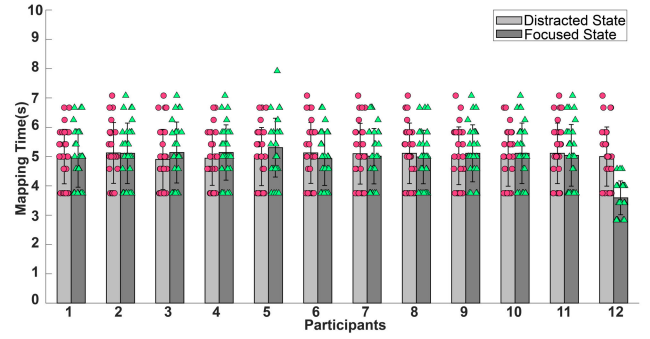
The information transfer matrix explores the knowledge about the signal transmitted to the participants and the knowledge of the signal received by the participant. This matrix has the advantage that it can be compared across different sets of stimuli. ITest (bits) is calculated by the formula given below:

$$IT_{est} (bits) = \sum_{i=1}^k \sum_{j=1}^N \frac{n_{ij}^T}{n^T} \log_2 \left( \frac{n_{ij} \times n}{n_i \times n_j} \right) \quad (2)$$

Eq. (2) is used widely for the computation of  $IT_{est}$  (bits) [23], [24], [25],  $k$  is number of stimulus alternatives,  $n$  is total number of trials,  $i$  and  $j$  are the indices for stimuli and responses respectively,  $n_{ij}$  is the number of trials when the stimulus  $i$  is reported as  $j$ ,  $n_i$  is the total number of trials that the stimulus  $i$  is presented, and  $n_j$  is the total number of trials that the user responds as  $j$ .

#### 1) MECHANICAL ENCODING STRATEGY

The data were first analyzed in terms of the correct responses reported by the participants as the function of the total trials participants attempted. The correct rate, and mean and standard deviation of the reaction time of the mechanical encoding strategy during the two testing blocks was analyzed (Table 3).

**FIGURE 6. Mapping time during the two experimental conditions for strategy 1. Red circle represents the mapping time for each trial during the distracted state and green triangle represents the mapping time for each trial during the focused state.**

To determine the effect of the external disturbance using a movie as a distraction on the reaction time and correct rate of the subjects, we conducted a paired t-test. The mean correct rate during the focused and the distracted test block was  $(95.22 \pm 0.04\%)(\text{mean} \pm S.D)$  and  $(90.55 \pm 0.05\%)(\text{mean} \pm S.D)$  respectively. The paired t-test indicated a significant difference in the performance of the subjects during focused and distracted state ( $t(1, 11) = -4.55, p < 0.05$ ). The mean reaction-time during the focused and the distracted test block was  $(9.60 \pm 2.47\%)(\text{mean} \pm S.D)$  and  $(11.09 \pm 2.79\%)(\text{mean} \pm S.D)$  respectively. The paired t-test indicated a significant difference in the reaction time of the subjects during focused and distracted state ( $t(1, 11) = 3.17, p < 0.05$ ). By decomposing the reaction time into the mapping time and the stimulus presentation time, we can further analyze the cognitive load for the two strategies required to decode the time information from the vibrotactile cues. The mapping time means the time utilized by the participant to estimate and map the encoding profile into the current time after the vibrotactile stimuli end. This time is calculated by subtracting the stimuli presentation time from the total time of the trial. The mean mapping time was  $(5.03 \pm 0.08)(\text{mean} \pm S.D)$  and  $(5.49 \pm 0.09)(\text{mean} \pm S.D)$  during the focused and distracted state respectively (Fig. 6). The RM-ANOVA results indicates a significant effect on mapping times ( $F(29, 660) = 12.17, p < 0.001$ ) between the trials however, the effect of the distraction was not significant ( $F(1, 660) = 0.21, p = 0.205$ ), indicating that the state of being distracted or focused does not significantly affect the mapping times. Additionally, the interaction between trials and distraction was not significant ( $F(29, 660) = 1.11, p = 0.319$ ), suggesting that the effect of trials is independent of the state of focus or distraction.

Results from the experiments are summarized in a 4-by-4 stimulus-response confusion matrix. The rows of the matrix corresponded to the stimulus (i.e. actual location of the activating tactator) and the columns refer to the response (i.e. reported location of the perceived tactator). Each cell contained the number of times a particular tactator was identified as being at a specific location. In the confusion matrix, the stimuli and the responses were represented by

**TABLE 6.** Confusion matrix for the strategy 1.

| Stimulus | Response |     |     |     | Sum  |
|----------|----------|-----|-----|-----|------|
|          | M1       | M2  | M3  | M4  |      |
| M1       | 386      | 3   | 1   | 10  | 400  |
| M2       | 8        | 360 | 17  | 0   | 385  |
| M3       | 2        | 18  | 467 | 14  | 501  |
| M4       | 23       | 0   | 2   | 129 | 154  |
| Sum      | 393      | 397 | 477 | 173 | 1440 |

the motors' locations instead of the other stimuli parameters such as the duration and the number of cues. The reason is that there were no confusions while counting the number of cues or differentiating the two levels of the durations (500ms and 1s). The only confusion arose in the localization of the motors. Table 4, given below is the stimulus-response confusion matrix of strategy 1. Table 4 lists the data pooled from all the 12 participants. Localization accuracy ranged from 96.5% (Motor-1) to 93.5% (Motor-2). However, localization accuracy is minimal for Motor-4 (83.7%). A visual inspection of the confusion matrix depicts that the maximum number of trials falls on the diagonal representing the correct responses. To observe the confusion across the columns and the rows the stimulus-response confusion matrices for each participant were combined. It can be observed from the data that the maximum confusion occurred in column-1 between Motor-1 and Motor-4, followed by Motor-2, Motor-3 and Motor-3, Motor-4.

Based on the confusion matrix presented below the  $IT_{est}$  values were calculated using eq. 2.  $IT_{est}$  values for each participant are listed in Table 7. The confusion matrix was constructed using the data of both conditions (focused & distracted state). We first calculated the  $IT_{est}$  values for each participant separately for the two different conditions and then all the  $IT_{est}$  values were averaged. The  $IT$  values determine that how many "bits" of information participants can distinguish from the set of encoded stimuli and indicate the maximum number of mapped digits can be identified without error. The maximum  $IT$  for 4 motors mounted on the wrist and the presented encoded vibrotactile tactons was 1.83bits. It means that there are 1.83 total possible bits of information to be transferred from the 4 motors that have to be identified. The calculation  $2^{IT}$  gives the maximum number of vibrotactile encoded tactons that can be correctly identified. Across all the participants the mean  $IT$  was  $(1.75 \pm 0.06 \text{ bits})(\text{mean} \pm s.d)$  for the wrist. This is interpreted as indicating that for this set of stimuli presented by 4-motors, on average 3.37 patterns could be correctly identified on the wrist.

## 2) DIGITAL ENCODING STRATEGY

The data were first analyzed in terms of the correct responses reported by the participants as the function of

the total trials participants attempted. The correct rate, and mean and standard deviation of the reaction time of the mechanical encoding strategy during the two testing blocks were analyzed (Table 5).

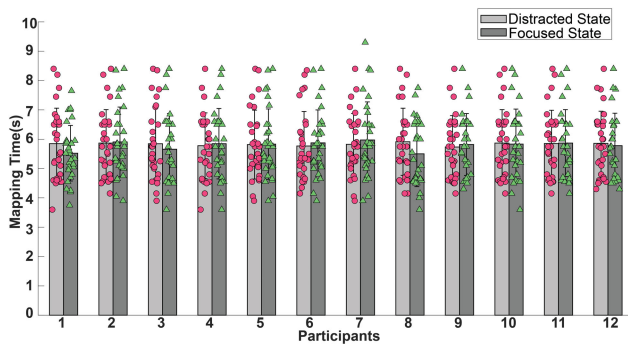
To determine the effect of the external disturbance using a movie as a distraction on the reaction time and correct rate of the subjects, we conducted a paired t-test. The mean correct rate during the focused and the distracted test blocks was  $(95.28 \pm 0.03\%)(\text{mean} \pm S.D)$  and  $(97.78 \pm 0.03\%)(\text{mean} \pm S.D)$  respectively. The paired t-test indicated that there was no effect of the distraction on the performance of the subjects ( $t(1, 11) = 1.518, p = 0.1572$ ). It is evidenced from the data presented above that higher accuracy almost 96 – 97% is achieved from the digital encoding pattern (Strategy-2). The resolution is also enhanced in this case as well up to 1 minute. The mean reaction-time during the focused and the distracted test block was  $(11.46 \pm 2.73\%)(\text{mean} \pm S.D)$  and  $(11.28 \pm 3.13\%)(\text{mean} \pm S.D)$  respectively. The paired t-test conducted on the reaction times also indicated no significant effect of distraction ( $t(1, 11) = 0.7826, p = 0.4504$ ). The mean mapping time for digital encoding strategy was  $(5.83 \pm 0.05)(\text{mean} \pm S.D)$  and  $(5.77 \pm 0.16)(\text{mean} \pm S.D)$  during the focused and distracted state respectively (Fig. 7). A two-way repeated measures ANOVA was conducted to examine the effects of the trial and state (Focused vs. Distracted) on mapping times. The analysis revealed a significant effect of trials on reaction times, ( $F(29, 660) = 12.09, p < 0.001$ ), suggesting that mapping times varied significantly across different trials. In contrast, the main effect of state (Focused vs. Distracted) was not significant, ( $F(1, 660) = 0.43, p = 0.510$ ), indicating no substantial difference in mapping times between the Focused and Distracted states. Additionally, there was no interaction between trials and state ( $F(29, 660) = 0.57, p = 0.966$ ), implying that the effect of trials on reaction times did not differ between the two states.

The results from experiment 2 are summarized by 10-by-10 stimulus-response confusion matrix. The row of the matrix corresponds to the stimulus delivered and the column of the matrix corresponds to the response. Each cell contains the number of times particular stimuli are responded to according to the stimuli. The  $IT_{est}$  values were also calculated using equation (2).

Table 6 lists the data pooled from all the participants during the focused and distracted state. A visual inspection of data pooled from all the participants in Table 5 depicts that most of the values are contained in the diagonal of the matrix. These values correspond to the success of the trials. The accuracy of the individual digits is as follows: Digit-0 (96.5%), Digit-1 (98.5%), Digit-2 (98.1%), Digit-3 (95.2%), Digit-4 (96.7%), Digit-5 (97.1%), Digit-6 (92.1%), Digit-7 (94.8%), Digit-8 (96.9%), and Digit-9 (93.5%). The  $IT_{est}$  values for this strategy were calculated first for all the participants and then averaged. The maximum  $IT$  for 2-motors mounted on the wrist presenting 10 vibrotactile encoded digits was 3.13 bits. It means that there are 3.13 total

**TABLE 7.** Results of the participants during the experiment 2 (C.R. = Correct Rate, R.T. = Mean of reaction time S.D. = standard deviation of reaction time).

| S.No.   | Focused |          |      | Distraacted |          |      |
|---------|---------|----------|------|-------------|----------|------|
|         | C.R.    | R.T. (s) | S.D. | C.R.        | R.T. (s) | S.D. |
| 1       | 100.00% | 9.13     | 1.76 | 100.00%     | 8.37     | 1.43 |
| 2       | 96.66%  | 10.28    | 2.12 | 100.00%     | 9.22     | 1.22 |
| 3       | 93.33%  | 11.42    | 2.46 | 100.00%     | 10.47    | 3.49 |
| 4       | 93.33%  | 10.62    | 3.12 | 96.66%      | 10.80    | 3.27 |
| 5       | 93.33%  | 9.54     | 1.85 | 100.00%     | 9.08     | 1.77 |
| 6       | 100.00% | 11.07    | 1.85 | 96.66%      | 10.23    | 2.01 |
| 7       | 93.33%  | 15.98    | 3.89 | 100.00%     | 16.40    | 5.48 |
| 8       | 100.00% | 11.25    | 4.19 | 90.00%      | 12.68    | 4.35 |
| 9       | 93.33%  | 15.05    | 5.07 | 90.00%      | 15.11    | 6.22 |
| 10      | 93.33%  | 10.36    | 2.35 | 100.00%     | 11.25    | 3.62 |
| 11      | 96.66%  | 10.04    | 1.87 | 100.00%     | 9.08     | 2.03 |
| 12      | 90.00%  | 12.81    | 2.28 | 100.00%     | 12.68    | 2.65 |
| Average | 95.28%  | 11.46    | 2.73 | 97.78%      | 11.28    | 3.13 |

**FIGURE 7.** Mapping time during the two experimental conditions for strategy 2. The red circle represents the mapping time for each trial during the distracted state and the green triangle represents the mapping time for each trial during the focused state.

possible bits of information to be transferred from the 2-motors presenting 10 tactions that have to be identified. The calculation  $2^{IT}$  gives the maximum number of vibrotactile encoded tac-tions that can be correctly identified. Across all the participants the mean IT was  $(3.08 \pm 0.04)(\text{mean} \pm \text{S.D})$  bits for the wrist. This is interpreted as indicating that for this set of stimuli presented by 2-motors, on average (8.46) patterns could be correctly identified on the wrist.  $IT_{est}$  values for each participant are listed in Table 7.

The confusion rate is calculated from the stimulus-response confusion matrix. The confusion rate is as follows: digit-0&2 (1.8%), digit-1&4 (1.2%), digit3&2 (2.3%), digit-4&7 (3.2%), digit-5&8 (1.8%), digit-7&4 (4%), digit-8&5 (3%) and digit-9&6 (6.4%). Therefore, maximum confusion appeared between digit-9&6. However, Participants reported that the maximum confusion for them was felt first between digit-0&2 and then digit-4&7. All the participants reported that the intensity of the motor's vibration was sufficient for the perception.

## D. DISCUSSION

The data presented above for the two strategies provide a significant insight into human psychophysics. In this

**TABLE 8.** Confusion matrix for the strategy 2 (S = Sum).

|   | Subject Responses |     |     |     |     |     |     |     |     |     |      |
|---|-------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|
|   | 0                 | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | S    |
| 0 | 476               | 4   | 9   | 0   | 4   | 0   | 0   | 0   | 0   | 0   | 493  |
| 1 | 1                 | 407 | 0   | 0   | 5   | 0   | 0   | 0   | 0   | 0   | 413  |
| 2 | 1                 | 3   | 429 | 0   | 0   | 4   | 0   | 0   | 0   | 0   | 437  |
| 3 | 0                 | 0   | 7   | 280 | 0   | 0   | 6   | 0   | 1   | 0   | 294  |
| 4 | 0                 | 0   | 0   | 0   | 237 | 0   | 0   | 8   | 0   | 0   | 245  |
| 5 | 0                 | 0   | 0   | 0   | 3   | 310 | 0   | 0   | 6   | 0   | 319  |
| 6 | 0                 | 0   | 0   | 7   | 0   | 1   | 140 | 0   | 4   | 0   | 152  |
| 7 | 0                 | 0   | 0   | 1   | 7   | 1   | 0   | 165 | 0   | 0   | 174  |
| 8 | 0                 | 0   | 0   | 0   | 0   | 6   | 0   | 0   | 191 | 0   | 197  |
| 9 | 0                 | 0   | 0   | 0   | 0   | 0   | 10  | 0   | 0   | 146 | 156  |
| S | 478               | 414 | 445 | 288 | 256 | 322 | 156 | 173 | 197 | 151 | 2880 |

**TABLE 9.** Information transfer (in Bits) from the two strategies.

| Participants | IT (Strategy 1) | IT (Strategy 2) |
|--------------|-----------------|-----------------|
| P1           | 1.83            | 3.13            |
| P2           | 1.81            | 3.12            |
| P3           | 1.78            | 3.07            |
| P4           | 1.72            | 3.04            |
| P5           | 1.76            | 3.08            |
| P6           | 1.79            | 3.11            |
| P7           | 1.76            | 3.09            |
| P8           | 1.75            | 3.08            |
| P9           | 1.64            | 2.97            |
| P10          | 1.73            | 3.09            |
| P11          | 1.78            | 3.11            |
| P12          | 1.63            | 3.08            |
| Average      | 1.75            | 3.08            |
| SD           | 0.06            | 0.04            |

section, we discussed the results obtained from the two encoding strategies separately. The first strategy utilizes the spatial encoding paradigm where each location represents one quarter, and the minutes and the hours are distinguished by the temporal pattern of the vibrotactile stimuli. The second (digital encoding) strategy utilizes the temporal pattern for each digit constructed by the duration variation and the number of cues.

## 1) MECHANICAL ENCODING STRATEGY

In the first encoding strategy, the dial of the mechanical wristwatch was mapped into four quarters using 4 vibration motors having 5 minutes of resolution. The idea was to utilize the spatial orientation of the motors to present each quarter uniquely. Two large durations were selected to present

hour and minute. The performance of this strategy was based on the localization performance of the participants. Gunther et al. suggested that the temporal duration of the stimuli determines the quality of sensation [70]. The results revealed that it is not significant to present information using more than two sites on the wrist in particular the thumb and the pinky region of the wrist. The results from the IT analysis indicate that by varying the duration of the vibrotactile cues and the location of the motors, participants were only able to identify 3-4 patterns reliably and attained an accuracy of almost 90% and 95% in focused and distracted environment respectively on the wrist.

The quality of the vibrotactile stimuli depends on the duration of the stimuli. Gunther et al. suggested that the temporal duration of the stimuli determines the quality of sensation. The two different durations were utilized to present hours and minutes to avoid confusion between hours and minutes [70]. It is assumed that for the abled users, this watch will be utilized in the scenario where the user is already familiar with the hours and needs to know just the minutes. In case the user requires only minutes and he/she presses the hour's presentation button by mistake, therefore, the different vibration patterns will help him/her distinguish the vibrotactile cues. Similar is the case for visually impaired users.

The watch was tested in two different conditions i.e. focused and distracted state. In the focused state, 95% correct rate was achieved whereas in the distracted state, the correct rate fell down to 90%. This study revealed that the environmental condition affects the performance of the watch. The reaction time achieved was 9.60s and 11.09s for the focused and the distracted state respectively. It is obviously a very slow response towards the watch for just perceiving time. However, this time can be improved by decreasing the stimuli duration and if we hypothesize that only minutes and hours were to be perceived this reaction time reduces to 4s or 5s which is still greater than the visual modality. However, due to technological constraints, we utilized the economical hardware for testing the proof of concept of this watch. In our pilot experimentation, we presented the hours and minutes consecutively instead of presenting them together. We found a very poor performance from that experimentation. The reason is the involvement of the working memory and selective attention. As suggested by Tyler et al. vibrotactile working memory shares common qualities with the other working memory systems [62]. Soros et al. also reported that working memory and attention are organized in partly overlapping neural circuits [71]. Therefore, we divided the information into two chunks and provided the control to the user. This method increased the performance remarkably but also increased the reaction time.

The selection of a suitable distance for the tactators decides the localization perception of the tactile displays. Azadi et al. evidenced that tactons were correctly identified on the forearm more frequently as compared to the hand [72]. Due

to the distance constraints, we suggested a circumferential layout of tactators around the wrist. Although we followed the guidelines suggested by Weinstein [18] while selecting a suitable distance for the placement of tactators around the wrist still the error was reported and recorded by the participants and experimenter respectively. In fact it is still very hard to conclude the exact distance for the placement of tactators as the wrist circumference varies from participant to participant. The maximum confusion was recorded in the Motor-1 and Motor-4 sites. The possible reason for this confusion is probably the misalignment of the wristband. Secondly, due to the variation in the wrist circumference, the distance between Motor-1 and Motor-4 sometimes becomes even smaller than the threshold of 38mm. The error reported mostly occurred by the motors located on the pinky and the thumb side of the wrist. This result is also justified by the result obtained by the experimentation proposed by Chen et al. evidenced that the dorsal and volar sides of the wrist are comparatively the most sensitive sides of the wrist [53]. But all these errors are mostly due to the technological constraints. This problem can be addressed in the future and can be solved by improving the technology.

## 2) DIGITAL ENCODING STRATEGY

In the second encoding strategy, the time is presented in the form of four digits having a 1-minute resolution. The digits were mapped into the vibrotactile cue to present time and tactons were created using three different durations. The reason for utilizing temporal pattern was to enhance the quality of the stimuli as the frequency was set to 220Hz the best frequency for the wrist [36]. Three durations were selected based on the pilot experimentation. Two spatial locations i.e. Dorsal and volar sides were selected based on the previous studies [30], [34]. The localization performance was enhanced and the accuracy was also enhanced by almost 96%. However, 4% error occurred due to the duration discrimination factor.

The overall performance of the participants in this experiment was comparatively better than the first experiment. The experimental results revealed that the correct rate achieved by digital encoding strategy in the focused and distracted states were 95.28%, and 96.39% respectively. This revealed the effectiveness of the spatial-temporal vibrotactile patterns for presenting time on the wrist.

In the first experiment, each side of the wrist was mapped with respect to each quarter of the mechanical watch dial. Therefore, spatial cue localization plays a significant role in the performance. However in this strategy, although still, the spatial cue localization is still important as the dorsal motor represents the Ten's digit and the volar motor represents the unit's digit the sites are reduced thus the localization performance is enhanced. As a result, overall performance during the experiment enhanced even in the distracted state.

The resolution in this experiment enhanced up to 1 minute and as a result the IT value also enhanced. All the participants responded positively about the difficulty level of the strategy

and duration discrimination. However, instead, we found confusion in the digits 1 and 4, 4 and 7, 6 and 9 and 5 and 8 as shown in Table 6. Although, the selected duration of vibrations was suitably long to be discriminated against utilizing the guidelines presented in the earlier studies even then could not be able to get 100% accuracy in a short training time. These findings suggest that tactons created using duration discrimination methodology still need more precise and quantitative conclusions. We utilized 3 different durations for designing suitable tactons and achieved an average of 95.22% and 97.78% accuracy in both states. In the context of spatial-temporal patterns, our study revealed that when utilizing the wrist, the best possible configuration with less hardware and less mental workload is to utilize the dorsal and volar sides of the wrist instead of the thumb and pinky sites. We still believe it is possible to present a large data set of information on the wrist by a minimum number of tactators with minimum cognitive load and less reaction time.

### 3) GENERAL DISCUSSIONS

The first and most important factor that plays a significant role in perceiving and manipulating information accurately in the case of wearable haptic devices is the cognitive load [73]. In the prototype, it was tried to develop a strategy that exhibits minimum cognitive load and presents maximum information in minimum time. Almost all the participants reported that both the strategies developed were easy to remember. The second factor that influences the accuracy is the design of the tactile pattern to present information. The more complex vibrotactile patterns can influence the training time and the mental workload.

The third and most challenging metric for the effective usage of the watch is reaction time. If it takes too long for a user to compute and decode the information, the device may become useless. The average reaction time was about 10s. The reaction time included the vibrotactile stimuli durations, the ISIs, and the computation and decoding time. While the mapping time was 4 to 5s on average, however, the vibrotactile stimuli time was a bit large. All the participants reported positively about the stimuli intensity and duration.

The cognitive processes include counting the no. of vibrotactile cues, short-term memory retrieval, and the mapping of vibrotactile patterns to time. We evidenced from the experimentation that it is significant to develop vibrotactile patterns and map the patterns to meaningful information. In the experimental setup, the participant responds verbally to the answer after every trial. The time to be presented is triggered by the buttons placed on the control box. However, these buttons could be replaced by touch sensors to make the design more ergonomic and more delicate.

### E. CONCLUSIONS AND FUTURE WORKS

We have presented filtering criteria for the design and evaluation of the industrial-scale vibrotactile devices for the wrist.

This is the first study that explored the fundamental concept of presenting quantitative information using vibrotactile stimuli on the wrist. Two encoding solutions are presented to achieve the goal of presenting time in the form of vibrotactile modality. Experimental results illustrated that the proposed strategy could be easily memorized by the users and achieve about 90–95% correct rate with only ten minutes of training.

With the increase of wearable haptic devices in real life, this study will provide a complete guideline for engineers to develop effective and robust wearable industrial-scale wrist-worn tactile devices for highly demanding environments. The proposed filtering criteria set is useful for selecting feasible solutions for customized information presentation tasks.

The results of the experiments revealed the limited ability of the human to process information with the vibrotactile modality in a very short time interval. Our future work will continue to explore better spatial-temporal patterns that could obtain higher correct rates, and require less reaction time and less training time. In the future, more robust and ergonomic hardware could be developed for further evaluation in real-world environments. In our future work, we also planned to incorporate various datasets to comprehensively assess the performance and applicability of the method across different contexts. This approach will not only strengthen the credibility of our results but also help identify potential limitations and areas for improvement. We developed this platform for the development of the haptic watch but this study opened numerous scientific questions. In this presented study, we did not measure the exact cognitive load but in the future, it is possible to use physiological measures such as EEG signals to evaluate the cognitive load while perceiving and decoding vibrotactile stimuli. These measures might provide a deep understanding of comparing and selecting different encoding strategies. Furthermore, it will be an interesting topic to explore the effects of multiple training sessions on the performance of the haptic watch and the retention effect of the vibrotactile encoding strategy.

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